

fraboulet.rosemarie@gmail.com
portfolio.romarin.tech
github.com/roma-rin
linkedin.com/in/r-mfraboulet
orcid.org/0009-0009-3021-0096

Skills.

Programming

Python · R · Shiny · Bash · LaTeX · HTML · CSS · JS

Pipelines & deployment

Docker · CI/CD · Version control (Git) · Cloud VMs · HPC cluster · Linux · Nextflow

Awards & grants.

- **2nd Thesis Prize** in Biology, Health and Environment, Fondation Université de Rennes, 2025
- **PhD grant**, Ligue Nationale Contre le Cancer, 2021-2024
- **Mobility grant**, Collège Doctoral de Bretagne, 2023

Talks & posters.

- **Talks:** JOBIM 2023 (remote), 24th Colloque de la Recherche, Ligue Contre le Cancer 2024 (Rennes), PhD Symposium 2024 (Rennes), "My thesis in 180 seconds" 2023 (Paris).
- **Posters:** CRCI2NA 2024 (Nantes), JOBIM 2022 (Rennes), JOBIM 2021 and EMBL The Non-Coding Genome 2021 (remote).

Beyond work.

- **Portfolio 2026:** personal website. github.com/roma-rin/portfolio
- **Sport:** road and trail running, strength training.

Languages.

French: native · **English:** fluent

Rose-Marie Fraboulet.

Computational Scientist · PhD

From academic research and experience abroad to a national data infrastructure, turning complex data into reliable results. Building pipelines and evaluating AI tools with engineers and partner labs, combining scientific rigor, fast learning and clear communication.

Experience.

2024 – now — **Computational Scientist**
Collecteur Analyseur de Données (CAD) · Paris, FR
National public infrastructure for storing and analyzing genomic health data.

Development, validation and integration of bioinformatics and AI tools for genomic and transcriptomic data from cancer patients, for care and research.

- **Rebuilt an existing immune profiling pipeline** in **Nextflow**, selecting the best published methods, on public data using **cloud VMs**.
- **Benchmarked machine learning** tumor primary site predictors on public data.
- Redesigned a regex parser into a reproducible, adaptable **Python** tool turning **sensitive clinical reports** into **structured data** on a secure **HPC cluster**.
- Across projects: packaged tools as **Docker** images in a container registry with **CI/CD** checks (lint, security scans), with DevOps teams and partner labs.

2021 – 2024 — **PhD Researcher**
Institut de Génétique et Développement de Rennes, GEO team · Rennes, FR
Supervision: Dr. Yuna Blum, Prof. Marie-Dominique Galibert

Led a **3-year research project** end to end: literature review, funding applications, coding, **data analysis**, **statistics** and scientific writing, to understand the circular RNA regulatory networks behind melanoma treatment resistance.

- Built and published Cirscan, an interactive **R Shiny** application that turns **multi-omics** data into **network visualizations**, on an HPC cluster.
- Presented results in **8 talks and posters**, for scientific and general audiences.
- **Supervised an intern** on a Nextflow pipeline detecting long-read circRNAs.

2023 — **Visiting Student Researcher** 4 months
Stanford University, Salzman team · Stanford, USA

- **Research collaboration** extending the PhD work on circular RNAs: identified their sequence variations using a k-mer approach.

2022 – 2023 — **Teaching Assistant**
Université de Rennes, Université de Paris

- **R classes** (~30 students): biostatistics (Bachelor, 36h), genomics (Master, 10h).

2021 — **M2 Intern** 6 months
Institut de Génétique et Développement de Rennes, GEO team · Rennes, FR

- Identified melanoma resistance RNA signatures on an HPC cluster.

2020 — **M1 Intern** 3 months
Ligue Nationale Contre le Cancer · remote

- Analyzed pancreatic tumor heterogeneity from scRNA-seq on an HPC cluster.

Education.

2021 – 2024 — **PhD in Bioinformatics**
Université de Rennes
Defended on Nov 28, 2024, with congratulations from the jury.

2019 – 2021 — **MSc in Bioinformatics for Health**
Université de Rennes

2019 — **Erasmus+ exchange** 6 months
University of Wrocław, PL

2016 – 2019 — **BSc in Life and Environment Sciences**
Université de Rennes

Publications.

- **Fraboulet** et al. Circscan: a shiny application to identify differentially active sponge mechanisms and visualize circRNA–miRNA–mRNA networks. *BMC Bioinformatics*, 2024. doi.org/10.1186/s12859-024-05668-y
- Si Ahmed, **Fraboulet** et al. circRNA sponge mechanisms implicated in resistance during melanoma treatment. *In review at NAR Cancer.*